STATEMENT OF REQUIREMENTS (SOR): ARMOR COMMAND (1998) MOBILE & STORE REBUILD

Document ID: SOR-AC98-2026-V3 (Autonomous Build, Procedural 3D PBR, AI Commander & Headless E2E QA Edition) | **Date:** October 22, 2026 | **Architect:** David Balasa

1. Executive Summary, Scope Refinement & Commercial IP Clean-Room Strategy

This Statement of Requirements (SOR Rev 3.0) defines the functional, visual, AI, architectural, and automated quality-assurance specifications to rebuild the 1998 military science-fiction real-time strategy (RTS) title Armor Command (originally developed by Ronin Entertainment, published by Ripcord Games, and designed by Edward Kilham) as a commercial, mobile-friendly web game and native store app (iOS App Store & Google Play via Capacitor).

To maximize autonomous development velocity and store reliability, Rev 3.0 locks in four core architectural mandates: (1) **100% Single-Player & Special AI Commander Focus** (eliminating real-time multiplayer netcode bloat); (2) **Procedural 3D PBR & Shader Asset Generation** in Three.js paired with zero-asset Web Audio synthesis; (3) **Commercial Clean-Room Rebranding** so the finished product can be legally sold as an original IP; and (4) a mandatory **Headless Browser & Agentic Full-Surface QA Swim Suite** that programmatically tests every screen, touch gesture, economy state machine, and AI battle before store submission.

System Dimension	1998 Original Baseline (Ronin / Ripcord)	2026 Rebuild Requirement (SOR Rev 3.0)
Orbital Reinforcement Loop	Unseen orbiting Seed Ship / Mothership beams units and supplies down to planetary **Landing Pads**; losing all pads halts reinforcement.	**Preserve 100%:** Maintain orbital drop-pad production queue, volumetric reentry drop beams, and multi-pad throughput scaling.
Dual-Resource & Tug Logistics	Surface **Ore** harvested by mobile Resource Miners; refined **Metal / Sentinel Metal** requires a Mining Station plus a physical **Tug** shuttling containers to a Landing Pad.	**Preserve & Visualize:** Highlight glowing Tug convoy lanes on the tactical HUD so raiding enemy Tugs and escorting friendly convoys are core mobile tactics.
Untethered Base & Mobile Defense	Buildings and turrets construct on any clear map coordinate without power-plant tether; turrets and depleted miners can be dismantled/sold or moved forward.	**Preserve 100%:** One-tap *Pack / Dismantle & Redeploy* radial command with animated fold-up sequences allows static defenses to leapfrog forward.
3D Graphics & Shader Pipeline	640×480 16-bit DirectDraw/Direct3D (armrcmdg.cfg); 150-poly flat-lit vehicles, stretched cliff UVs, and 2D bitmap explosion sprites.	**Autonomous Procedural 3D PBR:** 2,400-tri LOD0 compound meshes, industrial metallic PBR vs. biomechanical organic SSS shaders, triplanar terrain, and GPU instanced VFX.
Game Modes & Enemy AI	Campaign missions + 2–4 player LAN/Modem multiplayer requiring legacy DirectPlay.	**Single-Player Campaign + Special AI Commander Mode:** Strips multiplayer netcode; adds a 2-layer Utility AI Director with selectable Commander Personalities & Overpower survival waves.

System Dimension	1998 Original Baseline (Ronin / Ripcord)	2026 Rebuild Requirement (SOR Rev 3.0)
Pre-Release QA & Verification	Manual human playtesting; prone to config crashes (Old PC Gaming).	**Headless Browser & Agentic E2E Swim Suite:** Automated Chromium agents crawl 100% of screens, buttons, economy loops, and 100x-speed AI soak matches prior to release.

1.1 Commercial Resale Legality & Clean-Room IP Rules

Under **17 U.S.C. § 102(b)** and **U.S. Copyright Office Factsheet FL-108** (American Bar Association IP Law), game rules, resource systems, and functional mechanics are uncopyrightable, and any 1998 utility patent expired after 20 years (by 2018). However, under Tetris Holding, LLC v. Xio Interactive, Inc., visual "look and feel" trade dress, source code, and trademarks are protected. To ensure 100% commercial resale clearance on the App Store, Google Play, and Steam:

- **Zero Legacy Assets or Trademarks:** The build shall contain zero binary files, meshes, textures, or audio from the 1998 My Abandonware / Old Games Download archives, and shall replace the title *Armor Command* and faction names (*UTF*, *Vrass*, *Utopia*) with original commercial branding (e.g., *Orbital Drop Command*: **Terran Vanguard** vs. **Xeno-Symbiont Collective** over **Precursor Monolith** cores).
- **Original Expression via Upgraded 3D PBR & Custom Modes:** Because all 3D vehicle geometry, PBR shaders, mobile thumb-zone HUD layouts, Web Audio sound synthesis, and AI Commander modes are newly authored and visually distinct, the game passes the federal Abstraction-Filtration-Comparison (AFC) test as an original commercial work.

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2. Wrapper & Distribution Architecture (Single-Player Store Focus)

With real-time multiplayer removed, the distribution architecture simplifies to a **Single Modular TypeScript + Three.js Web Core** compiled to two targets: an instant-play **PWA / Android WebAPK** (for rapid web/headless testing and zero-tax direct distribution) and a **Capacitor v6+ Native Hybrid Wrapper** (for commercial Apple App Store and Google Play release).

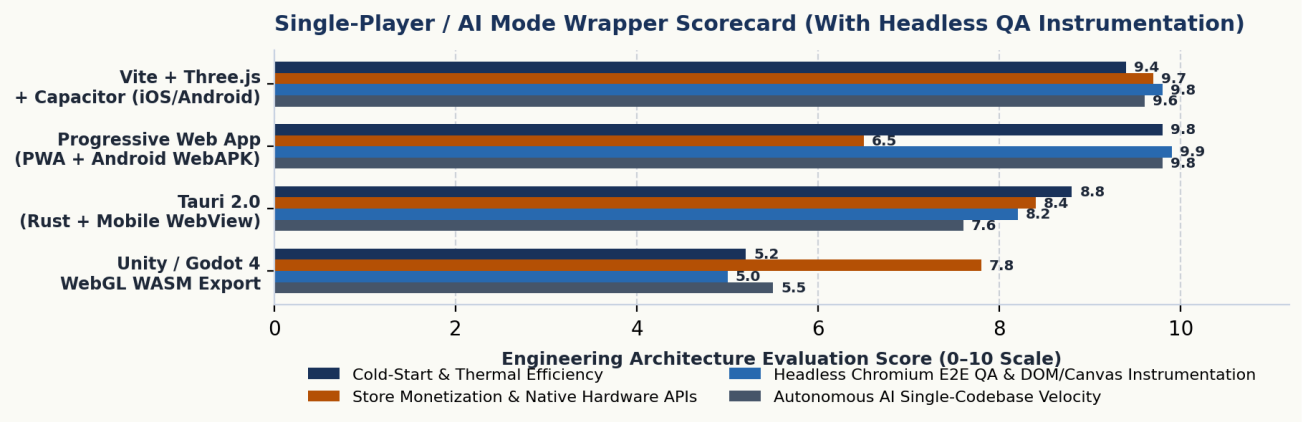


Figure 1: Engineering evaluation of single-player deployment wrappers across cold-start performance, store monetization, headless Chromium E2E testability, and autonomous AI development velocity.

Wrapper Target	Target Channels	Payload	Core Engineering & Store Advantages	Role in Build Pipeline
1. PWA + Android WebAPK (Primary Dev & Web Target)	Web URL, itch.io, Android Home Screen	2.2–3.2 MB	Zero install friction; 100% offline Service Worker cache; directly inspectable by headless Chromium E2E test runners at 100x simulation speed.	**Stage 1 Canonical Build:** Used for all development, headless browser QA swimming, and direct web play.
2. Capacitor v6+ Hybrid Shell (Primary Store Wrapper)	iOS App Store, Google Play Store	8.0–12.0 MB	Wraps the exact verified `dist/` web bundle into native WKWebView / Android WebView; unlocks native Haptics, True Fullscreen Immersive Mode, Orientation Lock, IAP/AdMob, and OS back-button capture.	**Stage 2 Commercial Package:** Compiled only after the web bundle passes 100% of Headless QA gates.

3. Core Gameplay Systems, 1998 Mechanical Fidelity & Special AI Mode

All core mechanics from the 1998 release—verified against Wikipedia, MobyGames, and gameplay telemetry (FirstPlays HD, WulfieZi, jhzh)—are preserved as deterministic state machines and extended with a dedicated **Special AI Commander Mode**.

3.1 Orbital Drop Pad Production & Dual-Resource Tug Logistics

- **REQ-PROD-01 (Orbital Landing Pad Anchor):** Units and building kits are queued in orbit and beamed down to terrestrial **Landing Pads** (UTF Mission 1). Losing all Landing Pads freezes orbital delivery until a replacement pad is constructed; each additional pad (up to 3) reduces drop cooldowns by 15% and serves as a selectable forward drop zone.
- **REQ-ECON-01 (Ore Miners & Depletion Auto-Dismantle):** Mobile **Resource Miners** extract general Ore directly from surface veins and automatically dismantle (refunding salvage credits) when a vein depletes (FirstPlays HD).
- **REQ-ECON-02 (Metal Mining Platform & Tug Convoys):** High-tier Metal / Precursor Metal requires a stationary **Mining Platform** plus an automated **Resource Tug** (VS Mission 2) that physically shuttles refined containers to the nearest Landing Pad—creating physical supply lines to raid and defend.
- **REQ-ECON-03 (Tech Levels 1–6 & Per-Unit Field Upgrades):** Metal unlocks global **Tech Levels 1–6** (VS Mission 11) and individual on-field vehicle upgrades (*"Scanning Vehicle — Upgrade Launched"*) that boost hull HP, weapon salvo damage, and sensor range.
- **REQ-BASE-01 (Untethered Placement & Pack/Relocate Defenses):** Structures and turrets place anywhere on clear terrain without power-grid tethers (Old PC Gaming) and can execute a 4-second **Dismantle / Relocate** order (VS Mission 3) to refund 70% cost or leapfrog forward.

3.2 All-Mechanical Unit Roster & Precursor Hazards

The roster is 100% mechanical (zero infantry) with decoupled hull movement and 360° turret tracking so combat units fire while moving:

- **12 Core Unit & Structure Archetypes:** Landing Pad, Resource Miner, Metal Mining Platform + Tug, Recon Scout, Light Assault Tank, Light/Heavy Missile Tank (VS Mission 8), Heavy Battle Tank,

Deployable Siege Artillery, Mobile Radar / Cave Probe (UTF Mission 9), Field Repair Vehicle, Aerial Gunship, and Deconstructible Defense/Radar Towers.

- **Precursor Battlefield Hazards:** Autonomous Precursor Towers, linked **Teleporter Portals**, hull-damaging **Charged Ground** conduits, and the central **Superweapon Obelisk** (which triggers a 60-second victory countdown once fed sufficient Precursor Metal, VS Mission 11).

3.3 Two-Layer Enemy AI & Special "Overpower / AI Director" Game Mode

Because units beam directly to Landing Pads and turrets require no base-crawling grid, the enemy AI never suffers from path-blocking base layout bugs. It operates via a two-layer deterministic controller:

- **Layer 1 — Strategic Utility-AI Director (0.5 Hz Tick):** Evaluates five normalized utility curves every 2.0 seconds:
  1. `U_econ`: Deploys Ore Miners or Metal Platforms + Tugs whenever active harvesters fall below target saturation.
  2. `U_tech`:Purchases Tech Levels 1–6 and triggers field upgrades on veteran tanks when Metal reserves exceed threshold.
  3. `U_def`: Places turrets along Tug supply lanes and packs/relocates backline turrets toward the active frontline.
  4. `U_recon`: Dispatches Scouts and Mobile Radar Vehicles to pierce player shroud and locate player Tugs/Radar Towers.
  5. `U_strike`: Forms combined-arms task forces (Heavy Tanks + Missile Tanks + Repair Vehicle) to execute coordinated strikes on player Tug convoys, Long-Range Radar, or Landing Pads.
- **Layer 2 — Tactical Squad Micro (20 Hz Simulation Tick):** Executes vector kiting (maintaining `0.85 × weaponRange` standoff distance while firing on the move), focus-fires the lowest-HP target in range, routes around Charged Ground hazards, and retreats damaged vehicles (< 25% HP) to the nearest Repair Vehicle.
- **Special Game Mode — "Overpower: Orbital Director":** Players face an adaptive AI Commander across procedural or curated maps with four selectable AI personalities driven by JSON weight profiles:
  - **The Convoy Raider:** High-speed hover scouts and gunships that relentlessly hunt player Resource Miners and Tugs.
  - **The Leapfrog Siege:** Advances a creeping line of deconstructible missile turrets, radar domes, and long-range artillery.
  - **The Xenomorphic Swarm:** Establishes 3 forward Landing Pads and floods the map with regenerating biomechanical assault tanks.
  - **The Obelisk Archon (Race Mode):** Prioritizes mining Precursor Metal and escorting Tugs to charge the central 60-second Superweapon.

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4. Mobile-First UX, Continuous 3-Tier Camera & Touch Grammar

- **Continuous 3-Tier Camera Spline (`z ∈ [0.0, 1.0]`):** Replaces the jarring 1998 camera snaps with a single smooth spring-damped zoom curve: **Tier 1 Satellite Strategic View** (`z ≤ 0.35`, 85° top-down with MSDF tactical icons and Tug route lines; selecting units here *never* forces a camera snap), **Tier 2 Tactical 3D View** (`0.35 < z ≤ 0.80`, 50° diagonal-down perspective), and **Tier 3 Action Chase View** (`z > 0.80`, 22° over-the-shoulder follow cam).

- **Touch Gesture Grammar:** 1-finger tap (Select / Move / Focus-Fire), 1-finger terrain drag (Inertial Pan), 2-finger pinch/twist (Zoom Spline + Orbit Yaw), **2-Finger Terrain Tap** (Attack-Move / Kite Waypoint), **Double-Tap Unit** (Select all on-screen units of same class), and **Long-Press Drag / Lasso Toggle** (Elastic box selection with haptic ticks).
- **Thumb-Zone Glassmorphic HUD:** Top resource/tech telemetry ribbon, bottom-left interactive Radar Minimap (with tap-to-jump camera), bottom-center quick command toggles (`Lasso`, `Atk-Move`, `Sat-View`, `Pause`), and bottom-right **Orbital Drop Pad Radial** + Squad Pills (1–5) + Context Action Radial (`Upgrade`, `Repair`, `Dismantle`). Supports both Landscape and Portrait layouts.

5. Autonomous Procedural 3D PBR Graphics Upgrade Pipeline

To eliminate multi-week external Blender asset bottlenecks while delivering a dramatic visual leap over the 1998 Direct3D baseline (Old PC Gaming), the engine uses an **Autonomous Procedural 3D PBR Geometry & Shader Pipeline** inside Three.js:

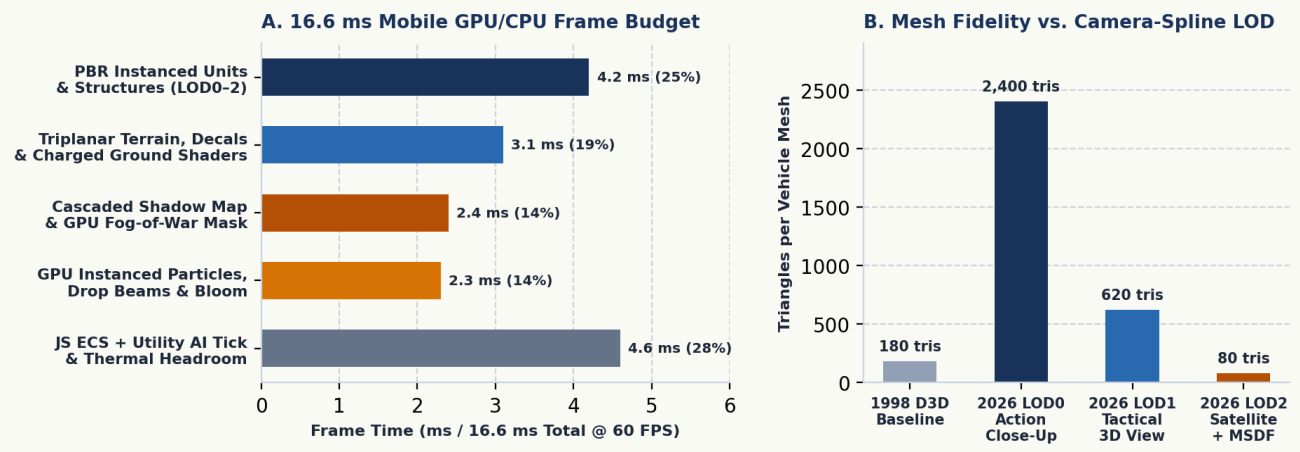


Figure 2: (A) Allocation of the 16.6 ms (60 FPS) mobile frame budget across WebGL2/WebGPU render passes; (B) Per-vehicle triangle scaling across the 1998 baseline vs. 2026 camera-spline LOD tiers.

- **Procedural Compound 3D Meshes & Custom Faction Shaders:** Vehicle hulls and independent 360° turrets are assembled programmatically from beveled compound primitives, extruded armor plates, and procedural normal/ORM canvas textures at startup:
  - **Vanguard (UTF) Hard-Surface PBR:** Brushed gunmetal metallic-roughness shaders (`metalness: 0.78`, `roughness: 0.34`), articulated tread/hover pods, vertex-shader barrel recoil, cyan emissive optics, and visible secondary armor/dual-barrel mesh nodes unlocked on field upgrade.
  - **Collective (Vrass) Biomechanical "Living Tank" SSS:** Iridescent chitin normal maps, subsurface-scattering (SSS) approximations, vertex-displaced organic breathing pulses, and magenta/amber bioluminescent veins (UTF Mission 2).
  - **Precursor (Sentinel) Monoliths & Voronoi Charged Ground:** Obsidian-and-gold monoliths, animated GLSL Voronoi plasma-arc shaders for **Charged Ground** conduits, and refractive portal shaders for **Teleporters** (VS Mission 11).
- **Triplanar Terrain, Ring-Buffer Battle Scars & GPU Fog of War:** 4-layer world-space triplanar terrain splatting (zero cliff UV stretching), a 512-entry GPU ring buffer for persistent tank tread tracks and blast craters, a 512×512 Gaussian-blurred GPU Fog-of-War mask with sweeping holographic Radar rings, volumetric orbital drop beams, and a 1-click **"1998 Retro CRT / Affine Mode"** toggle.

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6. Technical Architecture & Dual-State Memory Isolation

- **Strict React / Simulation Decoupling:** React 19 + **Zustand** render only low-frequency HUD overlays (updated at 10 Hz via transient reads), while all units, projectiles, tugs, flow-field vectors, and Three.js `InstancedMesh` matrices run inside a zero-allocation Vanilla TypeScript Structure-of-Arrays (SoA) singleton (`WorldSimulationEngine`) on a deterministic 20 Hz (`50 ms`) tick.
- **Mobile Hardware & Viewport Locks:** Enforces `100dvh / 100dvw`, `touch-action: none`, Android hardware back-button pause interception (`popstate` / Capacitor `backButton`), IndexedDB auto-snapshots on `visibilitychange`, and a Closed-Loop Dynamic Resolution Governor (DRS) that maintains 60 FPS under heavy combat load.

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7. Autonomous Headless Browser & Agentic E2E QA Swim Protocol ("Store-Readiness Gate")

No build may be packaged for the App Store, Google Play, or production release based on code inspection alone. Every build must pass an automated, multi-stage **Headless Chromium & Agentic E2E Swim Suite** (executed via Selenium WebDriver / Playwright in headless Chromium with WebGL enabled) that systematically exercises every screen, button, gesture, economy loop, and AI battle state.

7.1 Deterministic QA Telemetry & Control Bridge (`window.__QA_BRIDGE__`)

Because 3D WebGL canvases are opaque to standard DOM selectors, the engine must expose a global instrumentation hook (`window.__QA_BRIDGE__`) so headless browser agents can inspect internal 3D state, inject synthetic touch gestures, and fast-forward simulation ticks at 100x real-time speed:

```
// src/qa/qaBridge.ts — Exposed to Headless Chromium & E2E Test Agents
export interface QaInstrumentationBridge {
  // 1. Fast-Forward Deterministic Simulation (e.g., run 10 mins of game time in 2 seconds)
  stepTicks(tickCount: number): SimulationTelemetrySnapshot;
  // 2. Full State Inspection for Assertions
  getStateSnapshot(): {
    currentScreen: "TITLE" | "FACTION_SELECT" | "MISSION_SELECT" | "AI_MODE_SELECT" |
"IN_GAME" | "PAUSE" | "SETTINGS" | "VICTORY" | "DEFEAT";
    resources: { ore: number; metal: number; activeMiners: number; activeTugs: number };
    techLevel: 1 | 2 | 3 | 4 | 5 | 6;
    entities: Array<{ id: number; type: string; faction: string; x: number; z: number; hp:
number; upgraded: boolean; state: string }>;
    camera: { zoomSplineZ: number; tier: "SATELLITE" | "TACTICAL_3D" | "ACTION_CHASE";
retroMode: boolean };
    performance: { rollingFps: number; frameTimeMs: number; drawCalls: number; heapUsedMb:
number };
    runtimeErrors: string[];
  };
  // 3. Synthetic Touch & Command Injection
  dispatchCanvasGesture(type: "TAP" | "DOUBLE_TAP" | "TWO_FINGER_TAP" | "LASSO_DRAG" |
"PINCH_ZOOM", params: Record<string, number>): void;
  executeCommand(cmd: "DROP_UNIT" | "UPGRADE_TECH" | "SCAN_UPGRADE_UNIT" | "DISMANTLE_UNIT"
| "TOGGLE_RETRO", payload?: unknown): boolean;
}
```

7.2 Four-Stage Headless Browser & Agentic Swim Matrix

The automated Python/Chromium QA runner (`tests/e2e_headless_swim.py`) must execute all four verification stages across both mobile portrait (`393×852`), mobile landscape (`852×393`), and tablet

(`1024×768`) viewports with a **zero-error pass requirement**:

QA Swim Stage	Target Screens & Systems	Automated Headless Browser Actions & Hard Pass Assertions
Stage 1: Full-Screen & Modal UI Crawl	Every DOM screen, modal, drawer, radial button & viewport orientation	Agent visits **Title Screen → Faction Select → Campaign Select → Special AI Commander Mode Select (all 4 AI personas) → In-Game HUD → Drop Pad Radial → Tech Drawer → Squad Pills (1–5) → Pause Modal (including Android `popstate` back-button trigger) → Settings Modal (Audio + Retro '98 Toggle) → Victory/Defeat Screen**. Asserts `0` console/page errors and `0` clipped/overlapping DOM elements.
Stage 2: Economy & Combat State-Machine Swim	Landing Pads, Miners, Tugs, Tech 1–6, Upgrades, Dismantle & Combat	Agent drives a deterministic scenario via `__QA_BRIDGE__`: drops an Ore Miner → steps 400 ticks → verifies Ore increase & depletion auto-dismantle; builds Metal Platform + Tug → steps 300 ticks → verifies Tug container delivery & Metal increase; upgrades Tech 1→6; scans/upgrades a tank; packs/dismantles a turret (asserting exact 70% refund); verifies fire-on-the-move target destruction.
Stage 3: 100x Speed AI-vs-AI Soak & Stress Test	Utility AI Director, Flow-Field Pathfinding, Charged Ground & Memory	Runs a 20,000-tick (16.6-minute) 200-unit AI-vs-AI battle at 100x accelerated clock speed (~8 seconds wall time). Asserts: (a) zero `NaN` or out-of-bounds unit coordinates; (b) zero units permanently stuck on cliffs or Charged Ground; (c) heap memory delta < 5 MB across 5 consecutive match resets; (d) clean Victory/Defeat trigger via Pad destruction or 60s Superweapon countdown.
Stage 4: Multimodal Visual & Frame-Budget Audit	WebGL2/PBR Shaders, Camera Tiers 1–3, Retro Mode & Frame Time	Captures high-DPI PNG screenshots at `z = 0.20` (Satellite MSDF View), `z = 0.55` (Tactical 3D View), `z = 0.90` (Action Close-Up), and **1998 Retro CRT Mode**. Agent visually inspects rendered PNGs to verify shader contrast, HUD legibility, and `drawCalls ≤ 45` before clearing the build for store packaging.

8. Autonomous AI Build & Headless QA Execution Roadmap

With full autonomous design authority, procedural 3D PBR asset generation, no multiplayer overhead, and integrated headless browser QA, the end-to-end build and verification schedule compresses into five focused execution sprints (~2.5 to 3.5 hours total active execution):

Sprint	Est. Time	Milestone Title	Autonomous Build Scope & Headless QA Verification Gate
Sprint 1	20–30 min	**Core 3D Engine, Camera Spline & QA Bridge**	Scaffold `100dvh` viewport, procedural triplanar 3D terrain, continuous 3-Tier Camera Rig (`Satellite ↔ Tactical 3D ↔ Chase`), 20 Hz SoA ECS loop, and `window.__QA_BRIDGE__`. Verify with Headless Smoke Test.

Sprint	Est. Time	Milestone Title	Autonomous Build Scope & Headless QA Verification Gate
Sprint 2	35–45 min	**Procedural PBR Roster, Drop Pads & Tug Economy**	Build all 12 procedural 3D PBR vehicle/structure classes (Vanguard Hard-Surface vs. Collective Biomechanical SSS), Landing Pad drop beams, Ore Miners, Metal Station + Tug convoys, Tech 1–6, and Dismantle/Relocate. Run **Stage 2 Economy Swim**.
Sprint 3	25–35 min	**Utility AI Director & Special Game Modes**	Implement 2-layer Utility AI + squad kiting micro, 4 selectable AI Commander personalities, Campaign missions, Precursor Teleporters/Charged Ground, and the 60s Superweapon race. Run **Stage 3 20,000-Tick AI Soak Test**.
Sprint 4	25–35 min	**Thumb-Zone HUD, Touch Grammar & Web Audio**	Wire all screens (Title, Faction/Mode Select, HUD Radials, Tech Drawer, Pause, Settings, Retro '98 Toggle, Victory/Defeat) and procedural Web Audio SFX. Run **Stage 1 Full-Screen Crawl** across Portrait & Landscape viewports.
Sprint 5	20–30 min	**Full E2E Swim Audit & Store Packaging**	Execute complete 4-Stage Headless Chromium Swim Suite, inspect captured PNGs across all camera tiers and screens, fix any visual/layout edge cases, and package final PWA/WebAPK + Capacitor store build config.